

Master Build Spec: Intelligent Control Module (ICM)

1. Project Vision

The ICM is a code-first, resilient, sovereign business operating system for a UK-based vintage/antique retailer. It replaces brittle no-code/flat-file setups with a modular, scalable architecture that provides absolute data ownership and automated multi-channel sales operations.

2. Technology Stack

Database (Brain): Supabase (PostgreSQL) – Relational data integrity, secure auth, and persistent state.

Admin Dashboard (Office): Directus – Self-hosted, headless CMS providing a spreadsheet-style interface (grid/forms) for database management.

Automation Logic (Workers): Python – Custom scripts running in Docker containers (isolated and cross-platform compatible).

Agentic Orchestration: Claude Code (CLI) – Managing the code-first build, error-handling, and deployment.

Infrastructure Bridge: Docker & Docker Compose – Sandboxing all services to ensure OS independence.

3. The "Never-Break" Architectural Rules

Decoupled Architecture: Business logic must be separated from the UI. If a UI component breaks, the backend data remains safe and functional.

Atomic State Tracking: Use an "Outbox Pattern" for API integrations. No action is performed directly; tasks are queued in the database (`pending_tasks` table) for reliable, retrievable execution.

Data Sovereignty: The master inventory data is stored in the Supabase/PostgreSQL database. Third-party marketplaces (Etsy, Vinterior, Wix) are transient views, not the source of truth.

Validation-First Development: Every pipeline stage requires validation. Before committing to a full build, use "Feasibility Spikes" (small test scripts) to verify permissions and API connectivity.

Human-in-the-Loop (HITL): Automations shall prepare "Drafts" or "Pending Actions." Critical operations (moving money, publishing listings) require a human sanity check/confirmation.

4. Feature Checklist

Inventory Pipeline:

Unified Inventory Schema (SKU, Title, Status, Price, Provenance, Condition).

Automated metadata generation for vintage items (AI-assisted tagging).

Image Processing Pipeline:

Automated aspect ratio management (\$1:1\$ within \$4:3\$ canvas).

Batch processing via darktable-cli (using reference XMP files for consistent aesthetics).

Background removal and asset optimization for web.

Marketplace Integration:

Multi-channel listing (Etsy, Vinterior, Wix).

Asynchronous sync: Automated draft creation with human-review gates.

Marketplace API Watchdog: Error logging and automated retry loops.

Financial Ledger:

Starling Bank API integration (transaction monitoring).

Automated allocation engine (percentage-based splits for Ad Spend and Inventory Buying).

HMRC compliance engine (VAT Margin Scheme & Chattels boundary tracking).

5. Directory Topology (Local Storage)

01_ingest: Raw assets, source CSVs, unformatted manifests.

02_process: Intermediate vector/tokenized data fragments.

03_context: Sovereign knowledge base (markdown guidelines, voice guides, era signatures).

04_execution: Distilled business payloads ready for AI injection.

modules/: Self-contained code packages (image_processing, marketplace_integration, financial_logic, marketing_curation).